

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0395

Dixon Landing 230 kV
September 21, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0395. The Interconnection Customer proposes a 242.5 MW wind facility interconnecting at the Dixon Landing 230 kV substation in the Ohio area. The request was received on October 20, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 256 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,697,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Drsden to Conesville Sta 500kV line: 128.6% loading (2,422 MVA) following the outage of the Point Arena to Desden 500kV; DFAX = 5.1%.

For the loss of Piney Crk Project to Lions Rd 500kV, the Ganso to El Paso 500kV Line loads to 104.1% of its 2,322 MVA rating. Project DFAX: 14.9%.

The Orchard Road to Ft Humbug Substation 500kV is overloaded to 102.8% of its 3,640 MVA emergency rating for the loss of the Fort Humbug to W Campus 345kV. The project contributes 12.6% DFAX (30.6 MW) to this overload.

The Allegheny Energy Units 3 & 4 to Southeast Vicksburg 500kV Line is overloaded to 102.3% of its 2,511 MVA emergency rating for the loss of the Youngwood to Woodland Tap 345kV L. The project contributes 8.2% DFAX (19.9 MW) to this overload.

The Talenenergy Martins Creek Llc Harrisburg to Little Chute 345kV Line is overloaded to 101.9% of its 1,919 MVA emergency rating for the loss of the Coronet to Fort Humbug 500kV L. The project contributes 7.1% DFAX (17.1 MW) to this overload.

For the loss of Ft Humug to Orchard Road 500kV, the West Campus to Ft Humbug 345kV loads to 101.8% of its 2,615 MVA rating. Project DFAX: 8.4%.

The Scott to Shawville Station 345kV is overloaded to 101.7% of its 1,197 MVA emergency rating for the loss of the Orchard Rd to Shawville Station 500kV. The project contributes 5.5% DFAX (13.3 MW) to this overload.

Talenenergy Martins Creek Llc Williamsport to Montour Station 345kV: 100.5% loading (1,524 MVA) following the outage of the Orchard Road to Ft Humbug 500kV; DFAX = 5.5%.

The Shawville Sta 500-345 kV XFMR is overloaded to 100.4% of its 1,045 MVA emergency rating for the loss of the Carbon Ctr 500 kV/345 kV TX. The project contributes 5.1% DFAX (12.3 MW) to this overload.

For the loss of Hilltop to Talenenergy Martins Creek Llc Williamsport 500kV, the Hilltop Substation 500-345 kV TX loads to 100.1% of its 1,927 MVA rating. Project DFAX: 7.6%.

For the loss of Westwood Generation Llc to Coronet 500kV, the Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV TX loads to 100.0% of its 1,377 MVA rating. Project DFAX: 6.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dresden to Conesville Sta 500kV Line	Rebuild line with higher-capacity conductor	\$37,763,000	3% (\$1,225,000)
Ganso to El Paso 500kV	Reconductor line	\$34,507,000	6% (\$1,937,000)
Orchard Rd to Ft Humbug 500kV	Reconductor line	\$42,188,000	6% (\$2,331,000)
Allegheny Energy Units 3 4 & 5 to Southeast Vicksburg 500kV Line	Reconductor line	\$16,377,000	6% (\$897,000)
Talenenergy Martins Creek Llc Harrisburg to Little Chute 345kV Line	Reconductor line	\$10,307,000	6% (\$569,000)
Ft Humbug to W Campus 345kV line	Reconductor line	\$19,286,000	6% (\$1,065,000)
Shawville Sta to Scott 345kV	Reconductor line	\$37,529,000	6% (\$2,077,000)
Montour Sta to Talenenergy Martins Crk Llc Williamsport 345kV	Reconductor line	\$25,591,000	6% (\$1,414,000)
Shawville Sta 500-345 kV XFMR	Replace transformer with higher-rated unit	\$23,594,000	8% (\$1,830,000)
Hilltop 500 kV/345 kV Auto	Replace transformer with higher-rated unit	\$22,233,000	6% (\$1,230,000)
Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$38,443,000	6% (\$2,122,000)
	Total allocated to project		\$16,697,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.